

# Hochschule Ravensburg-Weingarten University of Applied Sciences



## Computer Vision Project Report : Task 2

### OBJECT DETECTION AND DEPTH ESTIMATION

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# Chapter 1

## Task Description

### 1.1 Introduction

In the context of autonomous driving, object detection and distance estimation is one of the key fields of research. Accordingly in this project we want to understand how we can use 2D detection to perform depth estimation to estimate the distance of the object. For this purpose we work on a subset of the KITTI dataset in a reduced format which consists of 20 scenes. For each respective scene, we have the intrinsic camera matrix and calculated 3D ground truth distances for each object in the scene as input. In this project we used the pretrained version of YOLOv8x for object detection and to extract the corresponding 2D bounding box coordinates. We only consider cars as the objects of scope in this project and neglect all other objects in the scene.

### 1.2 Depth estimation and Program logic

After detecting the vehicle using YOLO and generating the bounding box, we take the lower edge center point as the point of consideration for our distance estimation as shown in figure 1.1.



Figure 1.1: Illustrated point for distance estimation

Since we already have the camera intrinsic matrix ( $K$ ) as input. The next step is to project the point from given 2D image coordinates to 3D world coordinate system. For this purpose we make use of the below equation which describes the 3D line  $L$  that gets projected on to the image pixel.

$$L = \left\{ \begin{pmatrix} \lambda_1 \\ \lambda_2 \\ \lambda_3 \end{pmatrix} \in R^3; \quad \begin{pmatrix} \lambda_1 \\ \lambda_2 \\ \lambda_3 \end{pmatrix} = zK^{-1} \begin{pmatrix} z_1 \\ z_2 \\ 1 \end{pmatrix}, \quad \text{where } z \in R. \right\} \quad (1.1)$$

Since the camera is at a height of 1.65 metres from the driving plane, for the 3D projected line, we need to find the value of  $z$  that corresponds to the point where the 3D line intersects the driving plane as seen in figure. 1.2[1].

By multiplying  $K^{-1}$  and the homogenous pixel coordinates, we get a vector  $(\alpha_1, \alpha_2, \alpha_3)$ . Since we already know the Y-coordinate  $\lambda_2$  which is the height of the camera from the driving plane, we can get the value of  $z$  by the

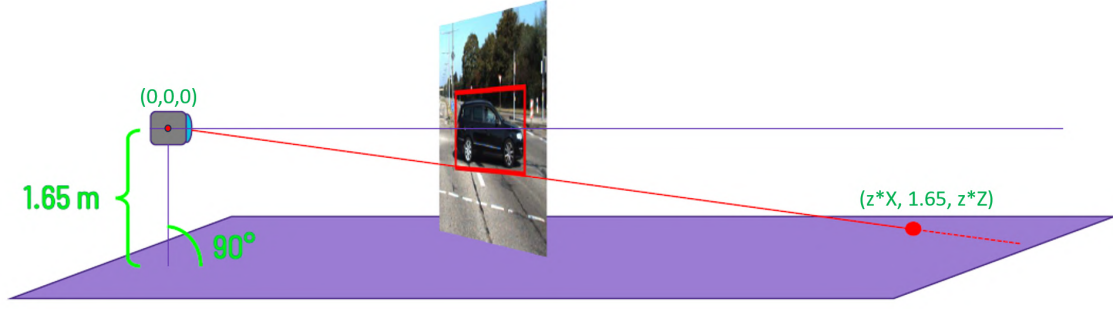


Figure 1.2: Projection of the point on the driving plane

equation,

$$z = \frac{\lambda_2}{\alpha_2}, \quad \text{where } \lambda_2 = 1.65\text{m} \quad (1.2)$$

From the obtained value of  $z$ , we can obtain the values of the remaining coordinates. Later we need to calculate the euclidean distance between the camera and the projected point. The formula for euclidean distance between two points is defined by the below formula.

$$Distance = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2} \quad \text{metres}, \quad (1.3)$$

For the evaluation of our estimation we compute the percentage error between estimated distance and ground truth distance using the below formula.

$$\text{Percentage Error} = \left( \frac{|\text{Estimated distance} - \text{Ground truth distance}|}{\text{Ground truth distance}} \right) \times 100\% \quad (1.4)$$

A flowchart illustrating the algorithm followed is shown in figure 1.5.

The images were represented in the below format as shown in figure 1.3 by labelling each car and showing the distance in the bottom frame of the image, as opposed to the format of representation given as reference, this is to avoid overcrowding of information as shown in below figure 1.4.

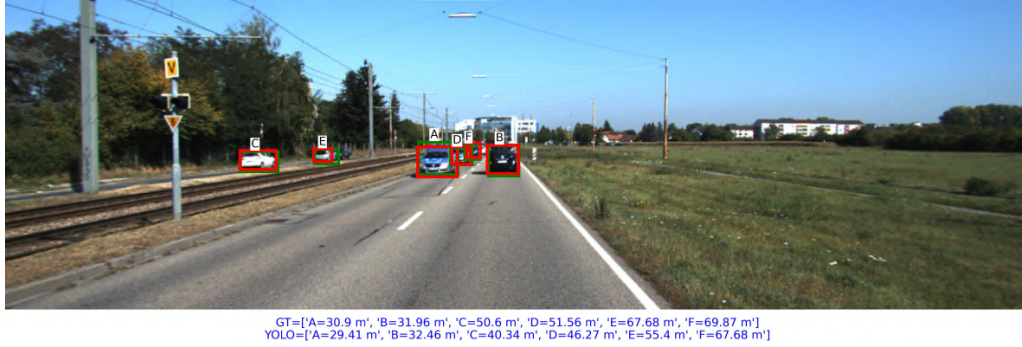


Figure 1.3: Scene 006059 in our format



Figure 1.4: Scene 006059 in given reference format

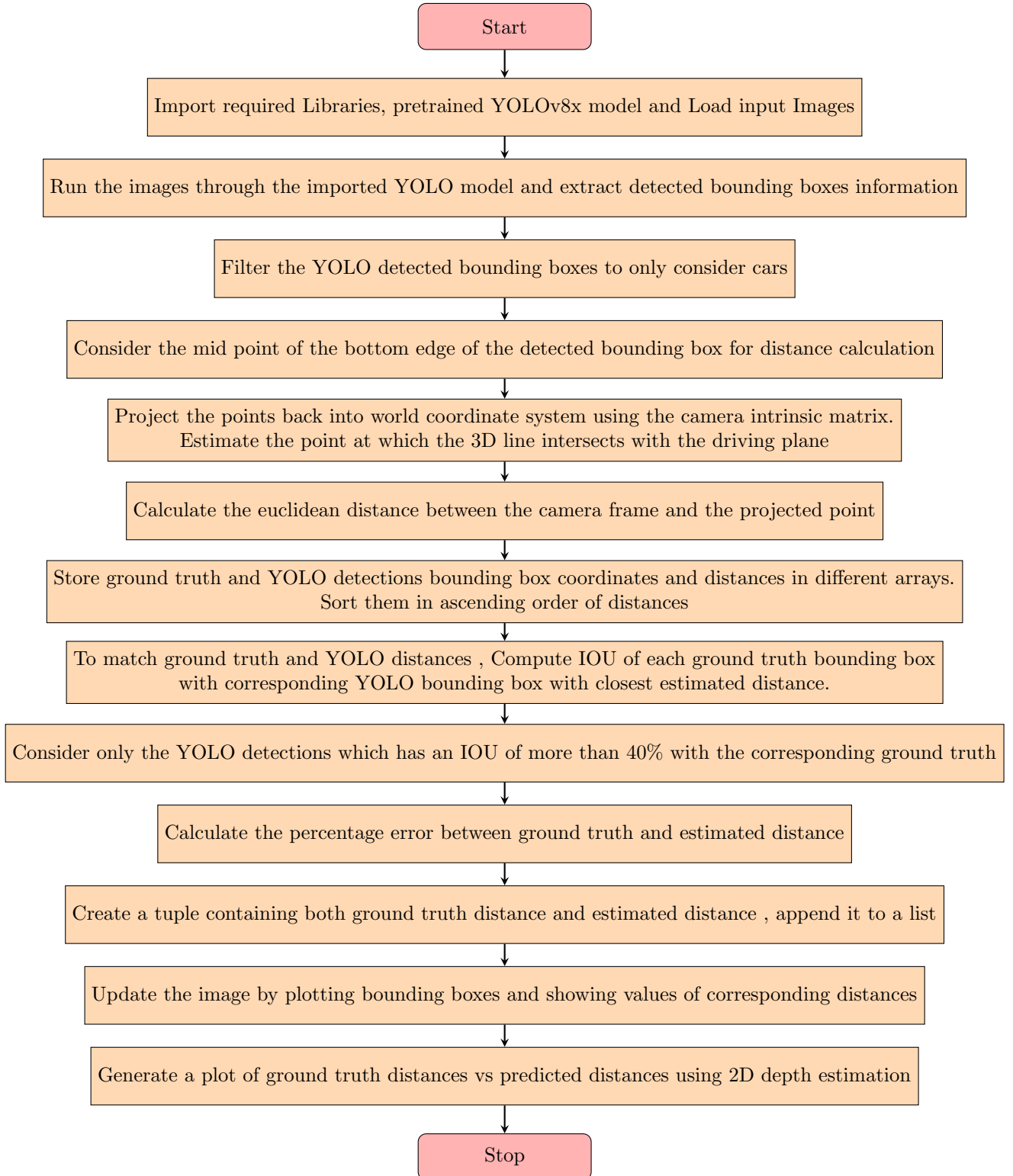


Figure 1.5: Flowchart illustrating the program algorithm

## Chapter 2

# Task Results and Observations

### 2.1 Observations

Since the ground truth was achieved using 3D point cloud data, we can see that with scenes with cars in close proximity as shown below, some cars given in ground truth are actually obstructed by the car behind in the camera, hence resulting in no detection by YOLO. Also sometimes, only parts of the car are visible and if a certain portion is visible, a detection is given by YOLO. However, since the bounding box generated will not match the actual ground truth bounding box, the calculated distance value will be inaccurate as seen below.

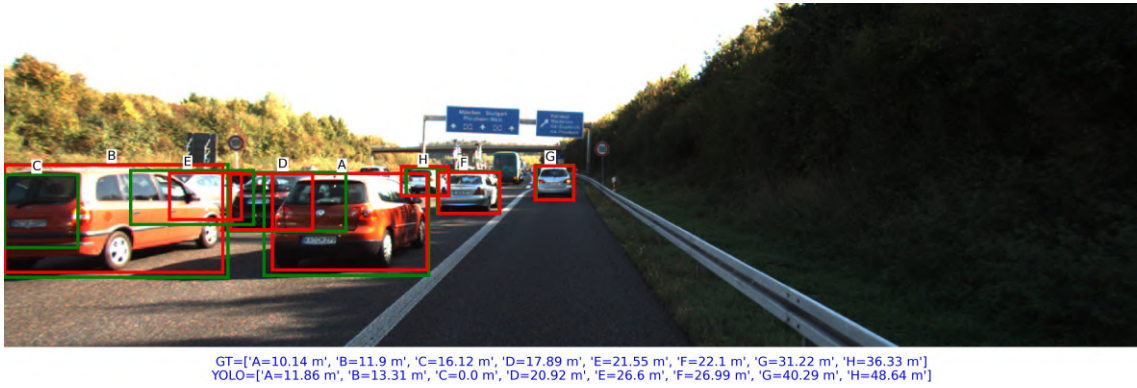


Figure 2.1: Scene 006312



Figure 2.2: Scene 006291

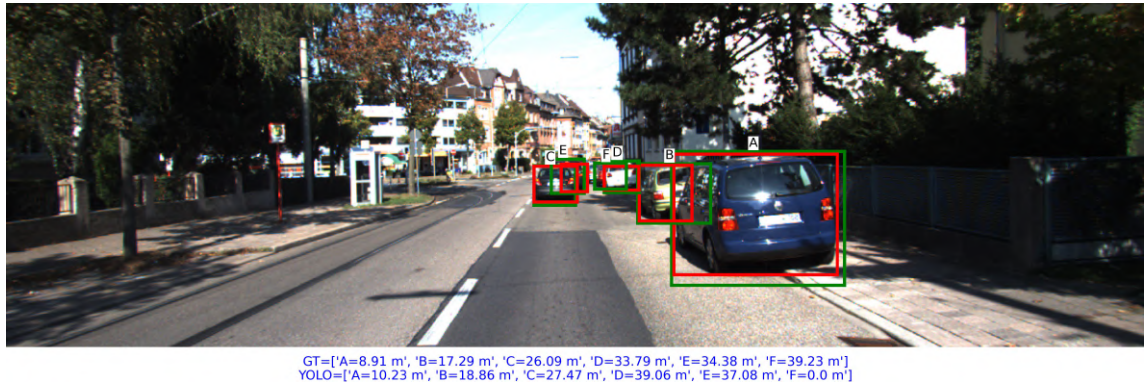


Figure 2.3: Scene 006253

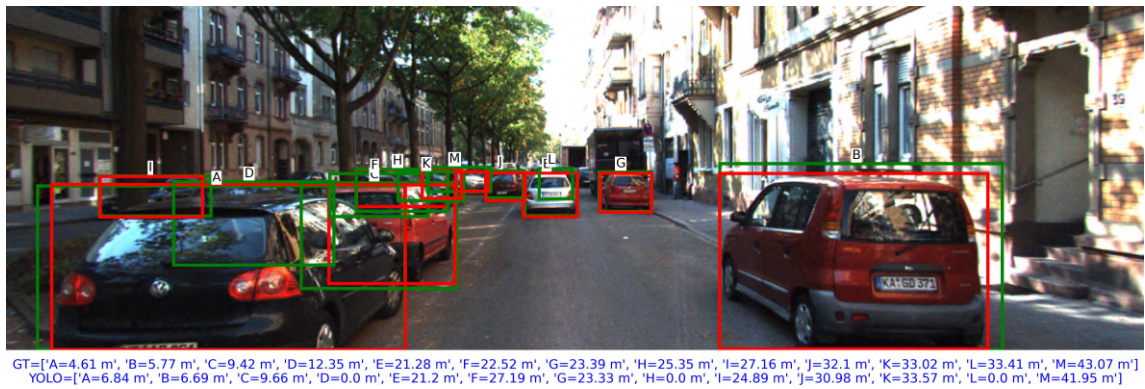


Figure 2.4: Scene 006211

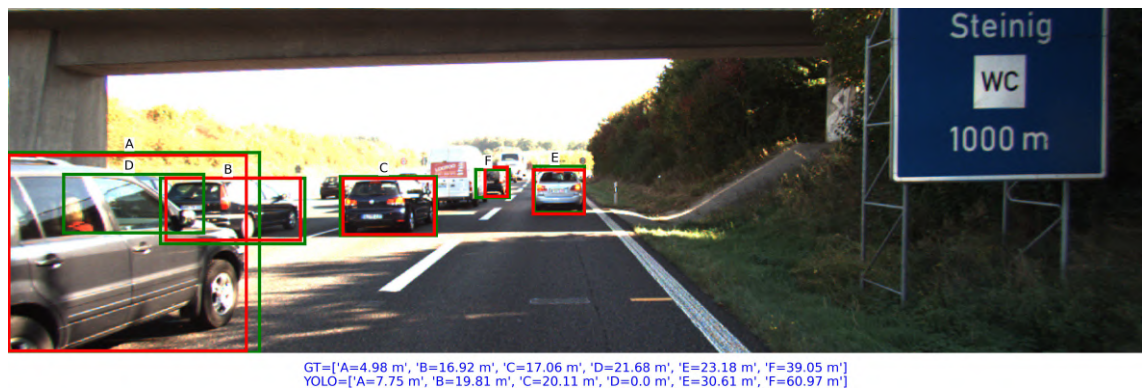


Figure 2.5: Scene 006048

In below scene 006310, we see that with increase in the road incline, the errors start to increase, resulting in a total outlier distance of 260 m for the farthest car as seen below. This error might be due the inclined points not correctly projecting on the driving plane.

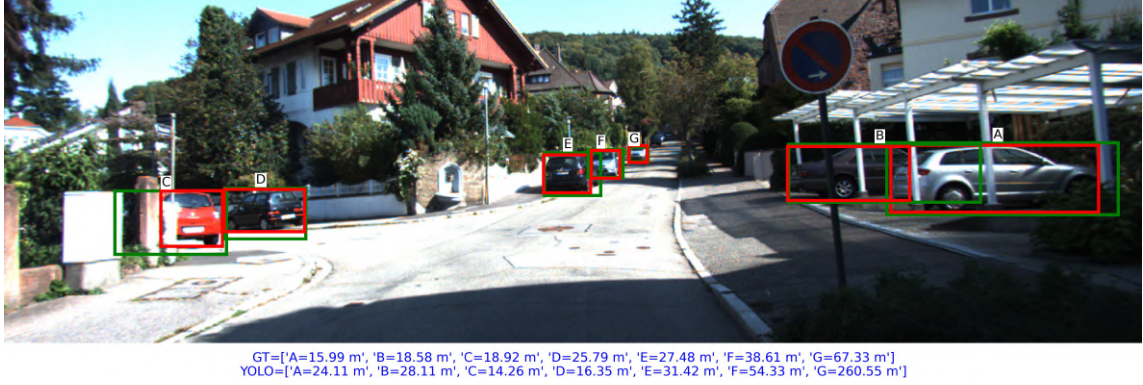


Figure 2.6: Scene 006310

There was no detections made and no ground truth data present in scenes 006121 and 006130 as there were no cars present as seen in below figures 2.7 and 2.8.



Figure 2.7: Scene 006121



Figure 2.8: Scene 006130

For cars which are too close to the camera, we see that the calculated error is high as seen in below figure 2.9. With the value of distance calculated for the car on the bottom right being 7.19 m compared to ground truth of 2.62 m. This is maybe due to the distance not being in the camera calibrated range and also due to the fact that the ground truth is calculated using the closest edge of the 3D bounding box, which in this case is not accurately estimated by YOLO since part of the car is hidden behind the camera.



Figure 2.9: Scene 006097

The results for the remaining scenes were mostly accurate as seen in below figures,



Figure 2.10: Scene 006042



Figure 2.11: Scene 006206

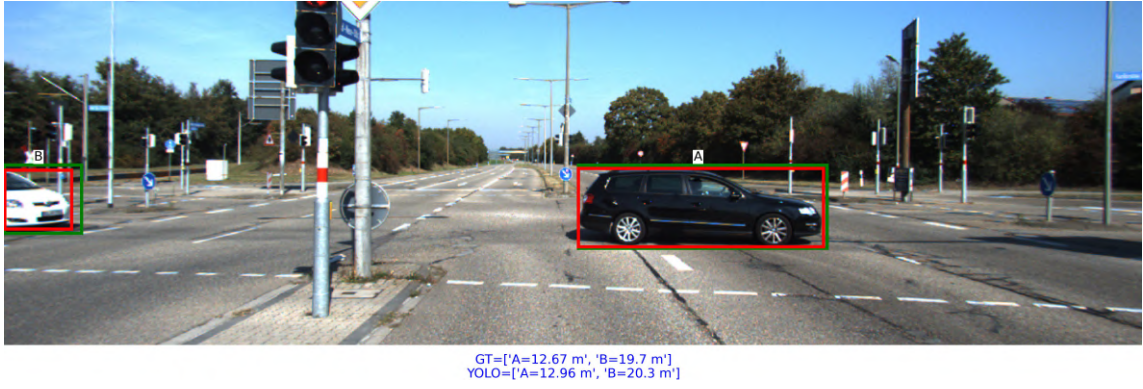


Figure 2.12: Scene 006227



Figure 2.13: Scene 006315



Figure 2.14: Scene 006374

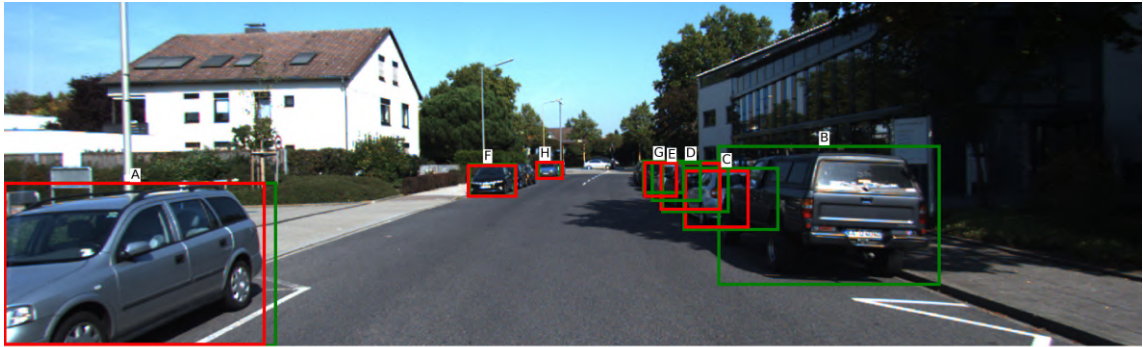


Figure 2.15: Scene 006329



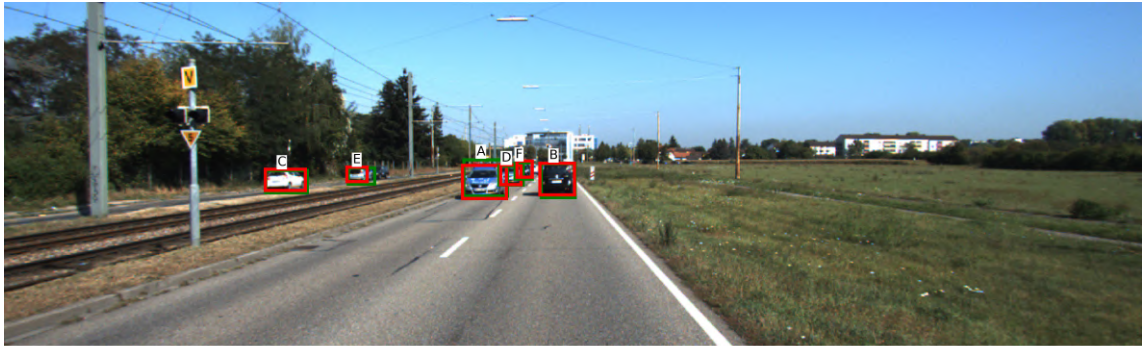
GT=['A=17.31 m', 'B=23.64 m', 'C=29.64 m', 'D=31.34 m', 'E=34.92 m']  
YOLO=['A=19.33 m', 'B=30.3 m', 'C=0.0 m', 'D=34.6 m', 'E=39.82 m']

Figure 2.16: Scene 006037



GT=['A=5.95 m', 'B=9.12 m', 'C=15.75 m', 'D=21.61 m', 'E=26.62 m', 'F=31.48 m', 'G=32.1 m', 'H=58.53 m']  
YOLO=['A=7.29 m', 'B=0.0 m', 'C=17.08 m', 'D=23.14 m', 'E=0.0 m', 'F=31.32 m', 'G=31.43 m', 'H=62.89 m']

Figure 2.17: Scene 006054



GT=['A=30.9 m', 'B=31.96 m', 'C=50.6 m', 'D=51.56 m', 'E=67.68 m', 'F=69.87 m']  
YOLO=['A=29.41 m', 'B=32.46 m', 'C=40.34 m', 'D=46.27 m', 'E=55.4 m', 'F=67.68 m']

Figure 2.18: Scene 006059



GT=['A=23.64 m', 'B=24.56 m', 'C=33.9 m', 'D=44.21 m', 'E=47.77 m']  
YOLO=['A=21.73 m', 'B=24.84 m', 'C=32.49 m', 'D=46.28 m', 'E=51.92 m']

Figure 2.19: Scene 006067



Figure 2.20: Scene 006098

Below plot 2.21 shows the overall comparison of predicted vs ground truth distances. The point which was an extreme outlier (predicted distance of 260 m) and the points which had no YOLO detections for corresponding ground truths were eliminated since they would result in an incorrect visualization. The mean percentage of error was calculated for the distance ranges of upto 10 metres, 10 metres to 35 metres and more than 35 metres. As also observed in the graph below, the mean percentage of error was very high for distances upto 10 metres with a value of around 63 %. For distances beyond 35 metres the mean percentage of error value was approximately 17 %. The distances range of 10 to 35 metres turned out to have the lowest mean percentage of error of around 12 %. Thus we can say that this distance range is the range of reliable distance prediction.

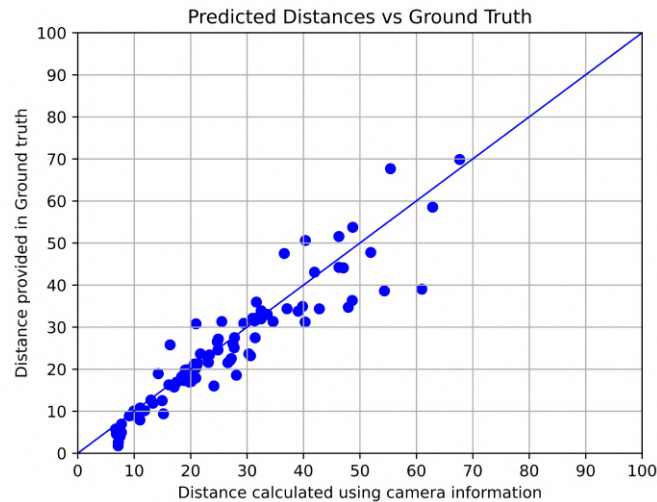


Figure 2.21: Plot of Predicted distances vs Ground truth distances

## Chapter 3

# Conclusion

In this project we were able to successfully perform depth estimation using 2D bounding boxes detected by YOLO and evaluate the accuracy by comparing it with the ground truth. From our evaluations we can make the following conclusions,

- The inaccuracy was very high for close distances upto 10 metres with a percentage of error of 63 %.
- The least percentage of error was found between the range of 10 metres to 35 metres which indicates the possible range of accurate camera calibration.
- The difference in 2D object detection using YOLO and object detection using 3D point cloud data is quite apparent in the scenes with a traffic jam where the camera fails to precisely capture vehicles whose significant portion is obstructed by another vehicle. This also led to incorrect depth estimation.
- For vehicles in close proximity, if the entire vehicle is not caught in the frame, the estimated distance will be inaccurate since the nearest point of the vehicle will be out of the camera's field of view.
- With increase in ground inclination, the inaccuracy of depth estimation increases significantly.

# Bibliography

- [1] Computer Vision - Prof. Dr. Stefan Elser. <https://elearning.rwu.de/course/view.php?id=2226>. [Online]. Date Accessed: June 12 ,2024.